STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-GOOD-ADMISSIBLE-SES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2982-2991 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 2993-3019, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 24 of queue_sdga.json (pb=1171, stmt 2982-2991, proof 2993-3019).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *et $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $0 \to \mathcal{P} \to \mathcal{P}' \to \mathcal{P}'' \to 0$ be an admissible short exact sequence of differential graded $\mathcal{A}$-modules. If two-out-of-three of these modules are good, so is the third.*

Proof. For condition (1) this is immediate as the sequence is a direct sum at the graded level. For condition (2) note that for any left differential graded $\mathcal{A}$-module, the sequence

$$0 \to \mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \to \mathcal{P}' \otimes_{\mathcal{A}} \mathcal{N} \to \mathcal{P}'' \otimes_{\mathcal{A}} \mathcal{N} \to 0$$

is an admissible short exact sequence of differential graded $\mathcal{O}$-modules (since forgetting the differential the tensor product is just taken in the category of graded modules). Hence if two out of three are exact as complexes of $\mathcal{O}$-modules, so is the third. Finally, the same argument shows that given a morphism $(f, f^\sharp) : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ of ringed topoi, a differential graded $\mathcal{O}'$-algebra $\mathcal{A}'$, and a map $\varphi : f^{-1}\mathcal{A} \to \mathcal{A}'$ of differential graded $f^{-1}\mathcal{O}$-algebras we have that

$$0 \to f^*\mathcal{P} \to f^*\mathcal{P}' \to f^*\mathcal{P}'' \to 0$$

is an admissible short exact sequence of differential graded $\mathcal{A}'$-modules and the same argument as above applies here. □

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